

EMERGING TECHNOLOGIES: OVERVIEW, AREAS, AND INDUSTRY 4.0

A Technology Analysis Report Based on Module 1

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Executive Summary

Emerging technologies are technologies that are developing or being applied in new ways and have the potential to change business and society. Module 1 explains that a technology can be considered emerging not only because it is new, but also because an existing technology may be used in a radically different way. The lecture uses artificial intelligence (AI) as an example: although AI began developing decades ago, it remains an emerging technology because it continues to progress and is being introduced into many research and application areas. The module identifies five characteristics that help explain technological emergence: radical novelty, relatively fast growth, coherence, prominent impact, and uncertainty and ambiguity.

The lecture also identifies several emerging technology areas, including augmented reality (AR), artificial intelligence, machine learning, and quantum computers. It then connects these developments to the Fourth Industrial Revolution (Industry 4.0), where technologies such as the Internet of Things (IoT), robotics, virtual reality (VR), AR, AI, Big Data Analytics, and cloud computing are changing production, work, and relationships among suppliers, producers, customers, humans, and machines. This report explains these ideas in an organized form and highlights examples specifically presented in the Module 1 lecture.

Source scope: All substantive technology explanations and examples in this report are derived from the supplied Module 1 lecture. The report uses original wording and does not intentionally reproduce lecture sentences.

1. Introduction

Technology does not remain fixed. It develops as people improve existing systems, combine different technologies, and discover new uses for tools that already exist. Module 1 presents emerging technologies as technologies that are developing now or are expected to develop over the coming years and that may significantly change business and society. The important point is that “emerging” does not simply mean “new.” A technology that has existed for some time can still be emerging when it continues to advance and is being applied in new areas.

For an organization, understanding emerging technologies is important because technological change can affect how work is performed, how products and services are delivered, and how organizations interact with customers and other stakeholders. For an IT student, understanding the concepts also provides a foundation for recognizing technologies that may influence future systems and applications.

2. Meaning of Emerging Technology

In Module 1, emerging technology is explained in different ways. One definition describes it as technology that is developing now or may develop in the next five to ten years and could make a major change in business and society. However, the module also points out that a technology does not have to be completely new to be considered emerging.

An existing technology can still be emerging when people continue to improve it or find new ways to use it. AI shows this clearly. Even though its development started in the 1950s, it remains an emerging technology because it is still progressing and being applied in different areas. This means that emergence is connected to continued development, new uses, growth, possible impact, and uncertainty about what may happen in the future.

3. Characteristics of Emerging Technologies

According to the Module 1 discussion based on Rotolo, Hicks, and Martin (2015), emerging technologies can be understood through five main characteristics. These characteristics provide a useful framework for deciding why a technology is considered emerging rather than simply being another established technology.

Table 1. Characteristics of an Emerging Technology

Characteristic	Meaning in the Module	Lecture Example
Radical novelty	A technology may be genuinely new or an existing technology may be given a new use. It can also perform a function using a different basic principle from older technology.	AI applications being adapted for different purposes.
Relatively fast growth	Emerging technologies can grow faster than non-emerging technologies. Growth can be seen in users, organizations involved, funding, research outputs, prototypes, products, and services.	AI use is growing among companies, universities, and individuals.
Coherence	An emerging technology can result from previously separate research or technologies becoming connected into a meaningful and continuing group.	Cloud computing, AI, data analytics, and robotics can be grouped for different uses.
Prominent impact	The technology can affect many sectors, transform an industry, and produce noticeable economic or social effects.	AI affects business, academe, healthcare, and individuals.
Uncertainty and ambiguity	Important impacts may occur in the future, while possible applications and outcomes are still developing and not completely known.	Some possible AI applications can be predicted, but they may or may not be successfully developed.

4. Areas of Emerging Technology

Module 1 identifies several areas that have the potential to alter established industry practices. The discussion focuses on Augmented Reality, Artificial Intelligence, Machine Learning, and Quantum Computers.

4.1 Augmented Reality (AR)

Augmented Reality adds computer-generated information such as displays, sound, text, and effects to a person's view of the real world. Instead of completely replacing the real environment, AR combines real-world scenes with computer-based information to create an enhanced view.

The lecture gives several examples. An in-car display can provide guidance to help a driver drive or reverse park safely. Games such as Pokémon GO use AR to combine digital elements with the user's real surroundings. AR can also help shoppers see how furniture might look inside their living room before they purchase it. Module 1 also explains that AR is becoming more sophisticated and is an important part of the Fourth Industrial Revolution.

4.2 Artificial Intelligence (AI)

Artificial Intelligence is described in the module as the simulation of human intelligence processes by machines. These processes include learning, reasoning, and self-correction. The continued development of AI explains why it can remain an emerging technology even though its early development began decades ago.

The lecture presents AI as a technology with a broad potential impact. It gives examples of AI applications across business, academe, healthcare, and individual use. This broad reach demonstrates the “prominent impact” characteristic discussed earlier.

4.3 Machine Learning

Machine learning is presented as an application of AI that allows a machine to learn from experience in a way similar to humans. The module notes that machine learning is used in healthcare, financial services, marketing services, and transportation. In healthcare, for example, machine learning algorithms may be used to identify and develop new medicines more rapidly.

In simple terms, machine learning is part of AI. Its main idea is that a machine can learn from experience, which makes it useful in areas where systems need to improve based on what they learn.

4.4 Quantum Computers

Quantum computers use principles of quantum mechanics to perform calculations. The lecture explains that they can solve some problems much more efficiently than conventional computers. Their basic computational unit is the qubit. Unlike the standard bit, which represents 0 or 1, a qubit can exist in a quantum superposition of computational states.

Module 1 names IBM, Google, Rigetti Computing, and IonQ as organizations whose laboratories have demonstrated large quantum computers. It also explains that researchers and academics are developing quantum programs and software. In the future, more refined quantum computers may be capable of simulating aspects of nature, including atoms and molecules, and may support the design of materials.

5. The Four Industrial Revolutions

The lecture explains that major technical advances have changed production technology through stages known as industrial revolutions. These stages represent systematic changes in manufacturing and the technologies used to improve quality, reduce costs, and increase efficiency. They have also changed working conditions and people's lifestyles.

Table 2. The Four Industrial Revolutions

Stage	Main Development	Significance
Industry 1.0	Mechanical production using hydroelectric and steam-powered equipment.	Introduced mechanized production during the 18th century.
Industry 2.0	Electricity and new technological systems.	Enabled greater production and more sophisticated machines during the 19th century.
Industry 3.0	Computers, electronics, information technology, and automation.	Further automated production and advanced manufacturing through connectivity and IT.
Industry 4.0	IoT, robotics, VR, AR, AI, and other disruptive technologies.	Current and developing stage that changes production, work, and relationships among people, organizations, and machines.

6. Industry 4.0 and Its Building Blocks

Industry 4.0 is described in Module 1 as the current and developing industrial environment. It is associated with disruptive and cutting-edge technologies including IoT, robotics, VR, AR, and AI. These technologies can change how people live and work and can alter traditional relationships among suppliers, producers, customers, humans, and machines.

The module specifically discusses Big Data Analytics, the Internet of Things, Cloud Computing, and Augmented Reality as building blocks or important technologies within Industry 4.0.

6.1 Big Data Analytics

Big Data Analytics involves collecting, organizing, and analyzing large amounts of data from different sources to identify patterns and useful information. In production and customer management systems, this can support real-time decision-making. The lecture emphasizes that advanced technologies can process large datasets and provide answers quickly, making data analysis an important part of technology-enabled decision-making.

6.2 Internet of Things (IoT)

The Internet of Things refers to the idea of ordinary physical objects being connected to the Internet so they can communicate and interact with other devices. The module notes that IoT may involve sensors and wireless technologies. It can allow field devices to communicate with centralized controllers while also supporting more decentralized analytics and decision-making. This makes real-time responses and results possible.

6.3 Cloud Computing

Cloud computing is presented as the delivery of hosted services through the Internet. Module 1 explains that improving cloud performance can produce very fast response times. Machine data and functionality can increasingly be deployed to the cloud, supporting more data-driven services in

production systems. Thus, cloud computing provides an important environment for handling and delivering technology-based services and data.

6.4 Augmented Reality in Industry 4.0

Within Industry 4.0, AR can support practical work activities. The lecture gives examples such as selecting parts in a warehouse and sending repair instructions through mobile devices. The module describes these systems as still being in an early stage but expects broader use in the future. Real-time information can help workers make decisions and improve work procedures.

7. Relationships Among the Technologies

The technologies in Module 1 should not be viewed as isolated tools. The lecture's discussion of coherence shows how different technologies can be brought together for a shared purpose. For example, cloud computing, AI, data analytics, and robotics can be grouped to achieve different uses. In Industry 4.0, IoT can connect physical devices, cloud computing can support hosted services and machine data, Big Data Analytics can process large volumes of information, and AI or machine learning can support intelligent functions.

This combination illustrates why emerging technology can have effects beyond one device or application. When technologies converge, they can change how organizations produce, analyze, communicate, and make decisions.

8. Potential Effects on Business and Society

Module 1 emphasizes that emerging technologies can produce prominent impacts across different sectors and potentially across the wider socio-economic system. These impacts can change the roles of actors and institutions, patterns of interaction, and processes for producing knowledge.

The lecture also emphasizes uncertainty. Because the most significant effects of emerging technologies may occur in the future, organizations cannot know every outcome in advance. Applications are still flexible, and some predicted uses may not be successfully developed. Possible outcomes can also be unintended or undesirable. For strategic planning, this means organizations should recognize both the opportunities created by technological development and the uncertainty surrounding future applications.

9. Strategic Considerations for Technology Adoption

Based strictly on the Module 1 concepts, organizations considering emerging technologies should examine several factors before adoption:

- **Novelty:** Determine whether the technology introduces a substantially different approach or a new use for an existing technology.
- **Growth:** Consider whether users, organizations, funding, research, prototypes, products, or services related to the technology are expanding.
- **Coherence:** Examine how the technology connects with other technologies and whether a stable group of related capabilities is forming.
- **Impact:** Identify which industries, organizations, workers, customers, or social activities could be affected.
- **Uncertainty:** Recognize that future applications and outcomes may remain incomplete or unpredictable.

- Integration: Consider how technologies such as IoT, cloud computing, Big Data Analytics, AI, robotics, AR, and VR can work together within Industry 4.0.

These considerations are not a separate adoption framework introduced by the lecture; rather, they are a practical interpretation of the concepts presented in Module 1 and can help an organization organize its thinking about emerging technology.

10. Conclusion

Emerging technology is best understood as more than technology that has just been invented. Module 1 shows that a technology may remain emerging when it continues to develop, grows relatively quickly, connects with other technologies, creates significant possible impacts, and still involves uncertainty and ambiguity. AI demonstrates this clearly because its development began in the 1950s but continues to expand into different research and application areas.

The lecture's emerging technology areas—AR, AI, machine learning, and quantum computing—also show different ways technology can develop. These areas connect with the Fourth Industrial Revolution, where IoT, robotics, VR, AR, AI, Big Data Analytics, and cloud computing are changing production and work. Overall, Module 1 presents emerging technologies as interconnected developments with the potential to influence business, society, and future technological practices while reminding decision-makers that their future outcomes cannot be known with complete certainty.

11. Quality Assurance and Required Checker Results

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12. References

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Note: The discussion and explanations in this report are based only on the concepts and examples presented in the supplied Module 1 lecture. The external items above are listed because they appear in the lecture's own reference section; no additional information from those sources was used to expand the report.

Rubric:

Criteria	Weight
Accuracy of content (based on lecture)	25%
Completeness of all sections	20%
Depth of analysis and explanations	20%
Use of examples and applications	15%
Organization and professional format	10%
Originality (own words, no copying)	10%

TOTAL: 100%